Growth and Decay

Introduction

Today, we will model population growth and radioactive decay using a differential equation we know well already:

$$\frac{dy}{dt} = ky.$$

The focus of today is not so much in solving the equation (we know how to do that already,) but rather in understanding why this particular equation models both our situations and in attaching terminology to various features of its solutions.

Review

Check that the solution of the initial value problem $\frac{dy}{dt} = ky$, $y(0) = y_0$ is $y(t) = y_0 e^{kt}$

Population Growth

Suppose that we have a small initial population P_0 on a large piece of land full of resources.

Explain why the following statement makes sense:

> "The rate of growth of the population is proportional to its current size".

Let's put some notation on this.

- Let the population at time t be $P(t)$. Then the rate of change of the population is given by _____.

- Write the statement "The rate of growth of the population is proportional to its current size" as an equation. Note: the word 'current' can be taken to mean 'at time t'.

You now have a differential equation for the population!

- We stated above that we start with a small initial population P_0. The word 'initial' here can be taken to mean 'at time $t = 0$'. Write this mathematically to add an initial condition to your DE.

- We already have the solution written above with different variables. Write it down now with these:

Assumptions vs Conclusions

We have done something new here. In previous work, our *base assumption* was that populations grow exponentially. We gave no reason for this, and took it as a given because we were told it's true. Here, our *conclusion* was that populations grow exponentially. Let's reexamine what assumptions led us there:

- What assumption did we make about the *rate of growth* of the population? Why is it more natural than 'the population grows exponentially'?

- Look carefully back again. What assumptions did we make about the *size* of the population? What two other assumptions were given? Why might changing those assumptions make it less obvious (or even untrue!) that population growth depends solely on the rate of reproduction?

Relative Growth Rate

If the population at time t is $P(t)$, then the rate of change of the population is _____. We're assuming populations grow, so this is the *growth rate*.

When population growth is described in textbooks or in the news, it's most often as a *relative growth rate*. For example, you may see it written that a population grows at 3% per year, or that the growth rate is 30 people per thousand per year.

Definition The *relative growth rate* of a population is the rate at which it grows relative to its current size. Mathematically, this is $\frac{\frac{dP}{dt}}{P}$, or $\frac{1}{P}\frac{dP}{dt}$.
Write down (again) the DE for exponential growth. Rearrange it to show that another way to say that a population grows exponentially is that its relative growth rate is constant.

If we don't assume that resources are unlimited, why might it make sense that relative growth rate decreases as population grows?

Later in the semester, we will study more sophisticated population growth models, in which letting relative growth rate decrease with population size instead of stay constant leads to more realistic results.

Putting Numbers on All This

1. Suppose that we start off with 20 people and that the population growth rate is $4\% = 0.04$ per year.

   (a) Write down the initial value problem corresponding to this situation and its solution.

   (b) How long does it take for the population to double from 20 people to 40?

   (c) How long does it take to double again from 40 to 80 people?

2. (a) Suppose now that we start off with 100 people instead of 20, how long would it take for the population to double to 200 people? Write out a new IVP and its solution first, then work with that.

   (b) Suppose that the population growth rate is $2\% = 0.02$. How long would the population take to double from a people to $2a$ people? Again, write out a new IVP and solution first.

3. As you can see, the amount of time it takes for a given population to double does not depend on the initial population, but only on the _____.

Definition The *doubling time* of an exponentially-growing population is the time it takes for the population to double.

Radioactive Decay

Given a large sample of a radioactive substance, the rate at which the substance decays is proportional to the amount of substance present, measured in number of atoms, or more usefully, in moles.

Suppose that we start with N_0 moles of a radioactive substance, and that the number of moles present at time t is given by $N(t)$.

Note that just like in the populations growth case, we have a rate of change that is proportional to the current quantity. But since we are now talking about a *decrease* or *decay*, we write a negative sign in front of the constant of proportionality:

$$\frac{dN}{dt} = _____, \; N(0) = N_0$$

This gives

$$N(t) = _____.$$

4. Suppose that the constant of proportionality for a given radioactive substance is $\lambda = 0.02$ per day. That is, it loses 2% of its amount each day. Calculate the *half-life* of this substance. That is, find the time it takes for the amount of substance to halve.

5. Thorium-234 has a half-life of 24 days.

   (a) If we start with 4 moles of Thorium-234, how long will it take until we only have one mole remaining? (Note: no solution of a DE needed here! How many half-lives must pass?)

   (b) Find the constant of proportionality λ for the decay model for Thorium-234. By approximately what percentage does the substance decay each day?

6. (a) Suppose that a radioactive substance decays exponentially with constant of proportionality λ. Find its half-life in terms of λ.

   (b) Suppose that a radioactive substance has half-life τ. Find the constant of proportionality for its exponential decay in terms of τ.